WORKED SOLUTION

STEP I 2006, Question 11

Mechanics • Elastic Collisions

> **STEP I 2006, Question 11**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/06-s1-q11

The key idea

We can't follow the collisions one by one: we don't even know the masses of $A_1, \ldots, A_{n-2}$. Instead we look only at the *final* state. Two facts constrain it. Momentum and kinetic energy are the same at the end as at the start. And once the collisions are over, no particle can be catching the one in front of it, so the final velocities increase (weakly) along the trough. Each part is then a short argument about which final states these two facts allow.

Solution

Setting up

We model the particles as points moving in a straight line without friction, with every collision perfectly elastic. Take the direction of A_0's initial motion as positive, and let v_k be the final velocity of A_k ($k = 0, 1, \ldots, n$). Write m_k for the mass of A_k, so $m_0 = m_{n-1} = m$ and $m_n = \lambda m$. Conservation gives

$$\sum_{k=0}^{n} m_k v_k = mu, \tag{M}$$

$$\sum_{k=0}^{n} m_k v_k^2 = mu^2. \tag{E}$$

The particles can't pass one another in the trough, so they stay in the order $A_0, A_1, \ldots, A_n$. If at the end $v_k > v_{k+1}$ for some k, then A_k would catch up with A_{k+1} and there would be another collision. So after all collisions

$$v_0 \leq v_1 \leq \cdots \leq v_n. \tag{O}$$

In particular, a particle moving forwards must have no stationary particle in front of it, and a particle moving backwards must have no stationary particle behind it.

Part (i)

Suppose only A_j moves at the end, with velocity v. Then (M) and (E) give $m_j v = mu$ and $m_j v^2 = mu^2$. Dividing, $v = u$, and then $m_j = m$.

Since $v = u > 0$, the ordering (O) needs every particle in front of A_j to have velocity at least u. They are all at rest, so there are none, and $A_j = A_n$. But A_n has mass $\lambda m \neq m$ because $\lambda > 1$. This is a contradiction, so exactly one particle can't be moving at the end.

Part (ii)

Suppose only A_{n-1} and A_n move, with velocities p and q. Dividing (M) and (E) by m,

$$p + \lambda q = u, \qquad p^2 + \lambda q^2 = u^2.$$

Substituting $p = u - \lambda q$ into the second equation gives $-2\lambda uq + \lambda^2 q^2 + \lambda q^2 = 0$, that is

$$\lambda q((1+\lambda)q - 2u) = 0.$$

We need $q \neq 0$ because A_n is moving, so

$$q = \frac{2u}{1+\lambda}, \qquad p = u - \frac{2\lambda u}{1+\lambda} = -\frac{(\lambda-1)u}{1+\lambda} < 0.$$

So A_{n-1} would be moving backwards. But $n \geq 2$, so there is a particle A_{n-2} behind it, and by assumption that particle is at rest. Then $v_{n-2} = 0 > p = v_{n-1}$, which contradicts (O): A_{n-1} would run into A_{n-2}. So this final state is impossible too.

Part (iii)

The main tool here is the following fact.

Lemma. At the end of the motion, some particle is moving backwards.

Proof. Suppose instead that $v_k \geq 0$ for every k. By (O), v_n is the largest velocity. Each term of (E) satisfies $m_k v_k^2 \leq v_n\, m_k v_k$, because $0 \leq v_k \leq v_n$ and $m_k > 0$. Adding these and using (M),

$$mu^2 = \sum m_k v_k^2 \leq v_n \sum m_k v_k = v_n\, mu, \qquad \text{so} \qquad v_n \geq u.$$

On the other hand, every term of (M) is non-negative, so the single term $\lambda m v_n$ is at most the total:

$$\lambda m v_n \leq mu, \qquad \text{so} \qquad v_n \leq \frac{u}{\lambda} < u,$$

using $\lambda > 1$ and $u > 0$. These two bounds contradict each other. □

Ask yourself The masses of $A_1, \ldots, A_{n-2}$ are never given. Why doesn't that matter?

The lemma only uses (M), (E), (O) and the mass of A_n, so it holds whatever the other masses are. That is why the question can be answered at all.

Now suppose only A_{n-2}, A_{n-1} and A_n move at the end.

If $n \geq 3$, there is a stationary particle A_{n-3} behind A_{n-2}. By (O), $v_{n-2} \geq v_{n-3} = 0$, and then $v_{n-1}, v_n \geq v_{n-2} \geq 0$. Every particle has velocity ≥ 0, which the lemma forbids.

If $n = 2$, the three particles are A_0, A_1, A_2, which is all of them, and the motion is completely determined. When A_0 hits A_1 (equal masses), the velocities are exchanged: A_0 stops and A_1 moves on with speed u. Then A_1 hits A_2, and by the calculation of part (ii) (with velocities u and 0 before the collision) it rebounds with velocity $-\frac{(\lambda-1)u}{\lambda+1}$ while A_2 moves off with $\frac{2u}{\lambda+1}$. Finally A_1 hits the stationary A_0, and again the equal masses exchange velocities. After that, A_0 moves backwards, A_1 is at rest and A_2 moves forwards, so (O) holds and there are no more collisions. A_1 ends at rest, so again it is impossible for all three to be moving.

Part (iv)

Suppose exactly two particles move at the end. By (M) the total momentum is $mu > 0$, so at least one of them moves forwards. By the lemma at least one moves backwards. So one, say A_i, has $v_i < 0$, and the other, A_j, has $v_j > 0$.

Ask yourself Which two particles can these be?

By (O), every particle behind A_i has velocity $\leq v_i < 0$, so it is moving. Only A_j is also moving, and it moves forwards, so nothing is behind A_i: $A_i = A_0$. Similarly, every particle in front of A_j has velocity $\geq v_j > 0$, so there are none and $A_j = A_n$.

So A_0 (mass m) and A_n (mass λm) are moving, with velocities $a < 0$ and $b > 0$. Equations (M) and (E) are

$$a + \lambda b = u, \qquad a^2 + \lambda b^2 = u^2,$$

exactly the equations of part (ii). Since $b \neq 0$,

$$b = \frac{2u}{\lambda+1}, \qquad a = -\frac{(\lambda-1)u}{\lambda+1}.$$

The two moving particles are therefore A_0, moving backwards with speed $\dfrac{(\lambda-1)u}{\lambda+1}$, and A_n, moving forwards with speed $\dfrac{2u}{\lambda+1}$.

This really can happen: the case $n = 2$ in part (iii) ends in exactly this state.

Checks

- With $\lambda = 3$ and $u = 4$, the speeds are 2 and 2. Then momentum is $m(-2) + 3m(2) = 4m$ and energy is $\frac{1}{2}m(4) + \frac{1}{2}(3m)(4) = 8m = \frac{1}{2}m(16)$, as it should be.
- As $\lambda \to 1$, A_0's final speed tends to 0 and A_n's to u: with equal masses everywhere, the velocity is just passed along the line, as in Newton's cradle.

- We simulated the full sequence of collisions for 60,000 random set-ups (n from 2 to 6, random masses for $A_1, \ldots, A_{n-2}$, random λ and spacings). A_0 always ended moving backwards, the final states in (i)–(iii) never occurred, and every time exactly two particles were moving they were A_0 and A_n with the speeds above.

Common slips

- Trying to track every collision. The intermediate masses are unknown, so only the final state can be analysed.
- Forgetting the ordering condition (O). Conservation alone allows the state in part (ii); it's the ordering that rules it out.
- Assuming A_{n-2} is not A_0. When $n = 2$ it is, and part (iii) needs a separate (short) argument.

Beyond STEP

The lemma is an inequality between two conserved quantities: with all velocities non-negative, the energy can't exceed (largest velocity) $\times$ (momentum). Arguments like this, which use conserved quantities and an ordering to rule out final states without solving the dynamics, are typical of the study of many-particle systems. A good next step is Kleppner and Kolenkow, *An Introduction to Mechanics*, which treats collisions, the centre-of-mass frame and conservation laws carefully.

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